

Effects of vegetation growth on soil microstructure and hydro-mechanical behaviour

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Literature studies are presenting counterposed effects of vegetation on soil hydraulic behaviour. Besides, root impacts at the micro-scale are rarely correlated to phenomenological observations due to the complexity of adopting up-scaling factors. In this study, the microstructural causes behind observed changes in vegetated compacted clayey sand's hydraulic and volume change behaviour were explored. Laboratory experiments were carried out on compacted samples seeded with *Cynodon dactylon*. Significant changes in soil water-saturated permeability, water retention and shrinkage upon drying were observed after root growth. Laboratory measurements were complemented by the quantification of root morphological features and observations at the micro-scale for different soil hydraulic states. X-ray scans and mercury intrusion porosimetry allowed to cover seven orders of magnitude of pore sizes, shedding light on soil fabric changes at the soil–root interface and the clay aggregate scale. The effects of roots on the soil pore size distribution consisted of Multiphysics phenomena: fissure generation, void clogging and soil aggregation due to roots' chemical interaction. Furthermore, a good correlation was found between the normalised volume of roots and the micro-pores volume. This new expression was included in a framework to predict soil water retention behaviour considering the aggregated structure and soil–root hydro-chemical interactions.

KEYWORDS: fabric/structure of soils; reinforced soils; unsaturated soils; permeability & pore-related properties; vegetation

INTRODUCTION

Vegetation is the fundamental component of the so-called nature-based solutions, sustainable works to mitigate natural hazards such as landslides and erosion processes. The effects of vegetation on mechanical soil behaviour have largely been explored. However, changes induced by roots on soil hydraulic behaviour are also relevant to model the soil–vegetation–atmosphere interaction properly, for example, during rainfall or drying periods (Tagarelli & Cotecchia, 2022; Ng *et al.*, 2023; Wang *et al.*, 2023). Many investigations have focused on the effect of plants on the hydraulic properties of soils (Lu *et al.*, 2020; Khalili *et al.*, 2022), observing different hydraulic responses depending on the soil type, its hydro-mechanical state during root growth, and the different plants investigated. Smethurst *et al.* (2012), Vergani & Graf (2016) and Li *et al.* (2016) found a general increase in water permeability within vegetated clayey soils; Ng *et al.* (2014) observed contrasting water retention responses in clayey soils compacted at different dry densities, Bordoloi *et al.* (2018) found an increase of water retention capacity in a vegetated

silty soil, while Pagano *et al.* (2019) did not observe any relevant effects of roots on a silty pyroclastic soil in loose conditions. Conversely, Capobianco *et al.* (2023) observed that a dense system of roots growing in a loose pyroclastic sand with silt contributed to an increase in retention capacity of the composite material. More recently, Tagarelli & Cotecchia (2022) found an increase in permeability and a decrease in the retention capacity of a vegetated clay. Soil dry density and water content impose constraints on roots' diameters, volumes and lengths, and their ability to open fissures and/or clog pores (Siegel-Issem *et al.*, 2005; Kolb *et al.*, 2017; Lucas *et al.*, 2019), with potential consequences for soil's hydraulic behaviour. Some studies on root effects at the pore scale have been produced: Carminati *et al.* (2013) and Liu *et al.* (2015) observed gaps opening at the root–soil interface due to soil and roots' concurrent shrinkage upon drying. Scholl *et al.* (2014) deduced the effects of roots on the pore size distributions (PSDs) of fine-grained soil by an inverse estimation, while Koebernick *et al.* (2017) performed X-ray micro-tomography focusing on a single soil–root interface. Lascrain *et al.* (2021) observed an increase in the total void ratio in the proximity of roots by X-ray micro-tomography. Bacq-Labreuil *et al.* (2018) observed increased porosity and connectivity in vegetated clayey soils and the opposite behaviour in sandy soils. Anselmucci *et al.* (2019) performed X-ray micro-tomography scans delayed in time, observing a general densification close to roots growing in loose sand and a general loosening when roots were left to grow in dense sand. Ng *et al.* (2020) observed that plant spacing also affected root growth and soil hydraulic behaviour.

Agronomic literature has focused on chemical interactions between roots and soils, indirectly affecting soil hydraulic behaviour (Ohu *et al.*, 1987; Watt *et al.*, 2006; Hinsinger *et al.*, 2009; Moradi *et al.*, 2011; Karup *et al.*, 2017). Indeed, polymeric substances produced by roots (mucilage) significantly modify the soil's chemo-physical properties by

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reducing the water surface tension, clogging pores and stabilising aggregates (Read *et al.*, 2003; Carminati *et al.*, 2010). The reduction of the water surface tension or the generation of a water-repellent surface inside a soil may decrease its retention capacity (Beckett *et al.*, 2016).

Within this knowledge, different methodologies have been adopted to infer/measure the hydraulic properties of vegetated soils, while the effects of roots on soil structure have rarely been quantified and related to plant morphological and physiological features (Dong *et al.*, 2023; Leung *et al.*, 2023). One example of modelling was provided by Ng *et al.* (2016), who proposed an empirical model dependent on the root volume normalised to soil bulk volume (i.e. root volume ratio, R_v) to formulate a reduction in void ratio due to the clogging of pores by roots, and to explain the increase of the soil's water retention capacity observed in their experiments. Ni *et al.* (2019) implemented the previous model with the impact of root decay. The identification of roots' features is relevant in 'vegetated soil mechanics' studies as: (a) roots constitute a soil phase different from solids, gases and fluids, which may alter the evaluation of void ratio and degree of saturation if not properly assessed; (b) given the practical complexities (i.e. representative elementary volumes) behind experimental studies involving vegetated soils, it is important to build models which could be reliable in predicting the behaviour of vegetated soil only by knowing that of the bare soil and the morpho-mechanical root traits: the latter are indeed intensively studied and implemented into databases.

Considering all this knowledge, there is still a research gap to fill and an increasing demand by practitioners of nature-based solutions relating to the correlations between microstructure, plant features and phenomenological observations on the hydraulic behaviour of vegetated soils. The aim of the present study is to explore the chemo-hydro-mechanical effects that roots have at the phenomenological scale and quantitatively estimate root effects on soil structure over an extended range of pore sizes, in order to contemplate multi-scale and multi-physical effects – from those due to the release of fluids from the roots to those mechanically generated by the roots growing to create spaces around solid soil particles. In addition, the root volume ratio will also be used as an up-scaling factor through its implementation in a soil constitutive framework.

SOIL PROPERTIES AND COMPACTION

The clayey sand used for testing was retrieved in the Llobregat River's delta (Barcelona, Spain) and characterised by Fraccica (2019). The resulting soil's fractions on sieving and the consistency limits of the fine fraction are presented in Table 1.

The soil was statically compacted in cylindrical pots (150 mm dia., 70 mm high) and in a permeameter (92 mm dia. and 221 mm high) by a hydraulic press at a displacement rate equal to 0.5 mm/min. The as-compacted state is indicated by point A in Fig. 1 (for cylindrical pots) and summarised in Table 1. After compaction, eight pots and one soil sample in the permeameter were seeded with *Cynodon dactylon* (plant spacing: 30 mm, four seeds on fixed points on the samples' surfaces, seeding density: 34 g/m²). All the samples were watered under unconfined conditions to a water content $w = 21\%$ and a matric suction around $s = 1$ kPa to allow plant development (point B in Fig. 1). Matric suction was monitored by a ceramic tip tensiometer (T5x, UMS, München, Germany). The potential soil volume change on wetting was checked by calliper (resolution 0.01 mm), but was not detected. The roots' growth lasted 3 months in the pots and 4 months in the permeameter until they reached the bottom of the samples. After establishment of the roots, samples were

Table 1. Main soil physical properties and as-compacted state

Soil properties/as-compacted state	Value
2 mm < soil fraction < 4.76 mm: %	9.1
75 µm < soil fraction < 2 mm: %	43.2
2 µm < soil fraction < 75 µm: %	39.3
Soil fraction < 2 µm: %	8.4
Liquid limit: %	29.5–34.4*
Plasticity index: %	9.6–13.5*
Mean grain size, d_{50} : mm	0.11
Density of solids, ρ_s : Mg/m ³	2.67
Gravimetric water content, w	0.15 ^{††}
Water ratio, $e_w = \rho_s w / \rho_w$	0.40 ^{††}
Void ratio, e	0.67 [†] –0.87 [‡]
Matric suction, s : kPa	40 [†] –8 [‡]
Max compaction vertical stress, $\sigma_{v,0}$: kPa	100 [†] –20 [‡]

*Oorthuis *et al.* (2018).

[†]As-compacted state in cylindrical pots.

[‡]As-compacted state in the permeameter.

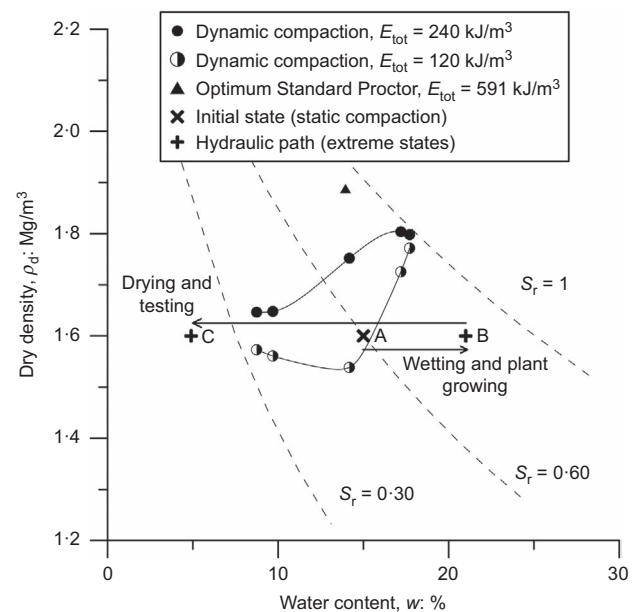


Fig. 1. As-compacted state for tested samples (point A), and low-energy compaction curves. Hydraulic paths followed after compaction: wetting (from A to B), plant growing (B) and drying (from B towards C) prior to test

subjected to air-drying, at null vertical stress, in a controlled temperature/relative humidity room (temperature, $T = 20^\circ\text{C}$, relative humidity (RH) = 50%) until reaching intermediate points between B and C in Fig. 1, at which measurements were carried out. Next, bare soil samples were prepared following the same path.

METHODS FOR PHENOMENOLOGICAL SCALE OBSERVATIONS

Water retention properties

Three vegetated pots and one fallow pot were used to determine the soil water retention properties. Matric and total suctions were assessed by a ceramic tip laboratory tensiometer, up to 0.20 MPa, and a chilled mirror dew point psychrometer (WP4, Decagon Devices, Pullman, WA, USA), from 0.50 MPa on. After each drying phase, which lasted 30 min, soil samples were placed for 24 h in glass desiccators inside a temperature/relative humidity-controlled room with natural light. Subsequently, specimens with roots were left

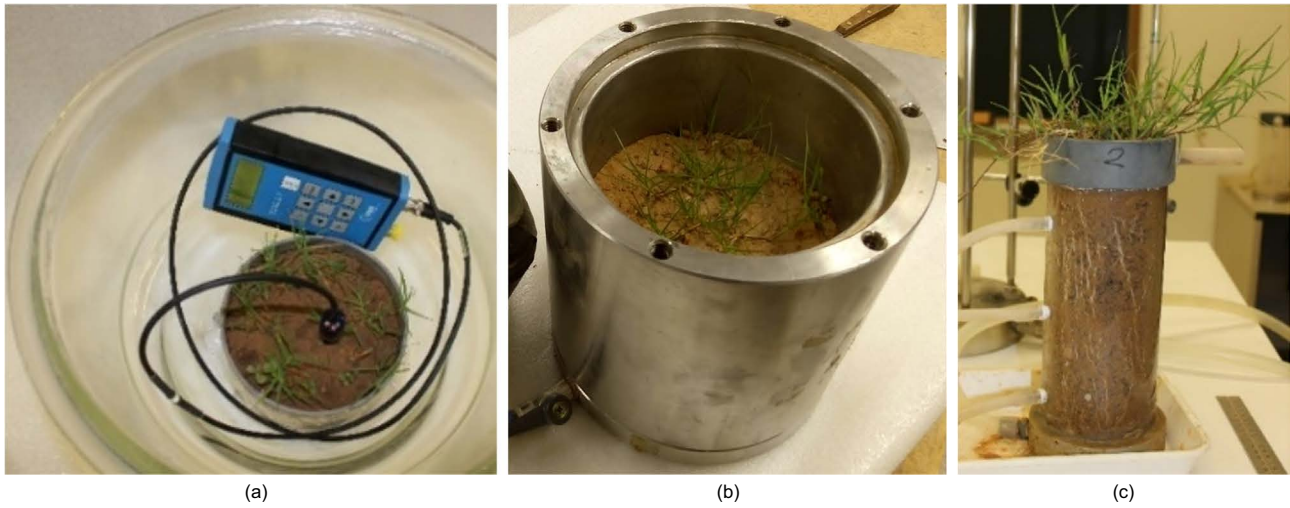


Fig. 2. Equipment for hydraulic measurements: (a) vegetated pot and tensiometer reading in the glass desiccator; (b) large cell oedometer; (c) permeameter

in the darkness for 2 h to induce stomatal closure (Snyder *et al.*, 2003) and to stop the plant's transpiration process completely (Howard *et al.*, 2009) to allow suction and water content equalisation, which are typically not homogeneously distributed due to plant roots' induced hydraulic gradients within the soil (Aylmore, 1993). After this period, the tensiometer was placed into the specimens up to a depth equal to their half height (35 mm), and measurements were performed inside the desiccator (Fig. 2(a)). The equalisation time of 2 h in the darkness was evaluated through a preliminary study, from which suction measurements independent of the proximity of the tensiometer to the individual plant were observed.

Water content during tensiometer measurements was evaluated by oven-drying (ASTM, 2019): two small soil samples of around 30 g each were extracted using a spatula, one from the upper and one from the lower surface of the pots. Visual inspection prevented the presence of roots inside them. This was done so as not to consider the water content included within the roots, which is no longer contributing to the generation of menisci in the soil and its water retention capacity. When the maximum measurable matric suction for the tensiometer was reached, soil samples were trimmed from the top and bottom of the pots and installed in the psychrometer.

Volume changes upon drying

Soil volume change was evaluated by paraffin tests (ASTM, 2018). Cubes of approximately 40 mm were trimmed out of fallow (five cubes) and vegetated pots (nine cubes) with an oil-lubricated spatula and at different suctions and water contents. Suctions and water contents were achieved with the same procedure explained for the soil water retention properties. The cubes were trimmed out from the bottom parts of 70 mm high vegetated soil samples. The cubes were first weighed (with a resolution of 0.1 mg) without paraffin; in the case of vegetated soil, the plant's aerial part (shoots) was cut with scissors before weighing. Disturbance of the soil cubes was avoided by visual inspection. Samples were then immersed in liquid paraffin at a temperature of 60°C. Once the paraffin had solidified, cubes were weighed again in above-water and under-water (hydrostatic weighing) conditions. These three measurements allowed the cubes' total volume V_{tot} to be calculated (Fig. 3).

In the vegetated soil cubes, the computation of the total volume V_{tot} by the paraffin method included the volume of roots (Fig. 3). So, after submerged weighing, vegetated soil cubes were destroyed to recover the roots present inside them.

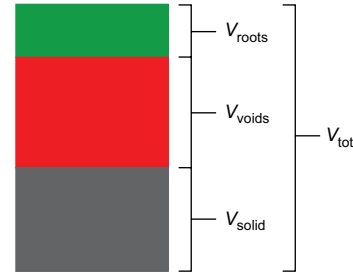


Fig. 3. Schematic diagram of the volumes considered in the root-permeated soil (re-adapted from Wood *et al.* (2016))

Then, the dry mass $M_{d,s}$ and the gravimetric water content of the soil deprived of its roots were estimated by oven check. Finally, root volumes (V_{roots}) were measured by pycnometry and image analyses, as described in the next section. Then, the volume of the voids was calculated as the difference:

$$V_{\text{voids}} = V_{\text{tot}} - V_{\text{roots}} - \frac{M_{d,s}}{\rho_s} \quad (1)$$

considering the density of solids ρ_s indicated in Table 1. Hence, the void ratio was computed according to its classical definition ($e = V_{\text{voids}}/V_{\text{solid}}$).

Root volumes by pycnometry and image analyses

Roots extracted from soil samples were placed in a pycnometer (Fig. 4(a)) filled with 100 ml of distilled water. The system was vibrated for 30 min to let the air bubbles come up, and then, the water raised beyond the control line due to the immersion of the roots was extracted with a pipette with a resolution of 0.01 ml and assumed as the roots' volume.

The commercial software WinRHIZO (Regent Instruments Inc., Quebec, Canada) allowed a redundant estimation of the root volumes. The software captures and binarises images (Fig. 4(b)) to evaluate root diameters (as projected area per length) and total length.

Given the root length (L_i) for each root diameter class (ϕ_i), the total root volume (V_{roots}) is then estimated by summing up the partial values ($V_{\text{roots}, i}$), according to

$$V_{\text{roots}} = \sum_i V_{\text{roots}, i} = \sum_i \pi \frac{\phi_i^2}{4} L_i \quad (2)$$

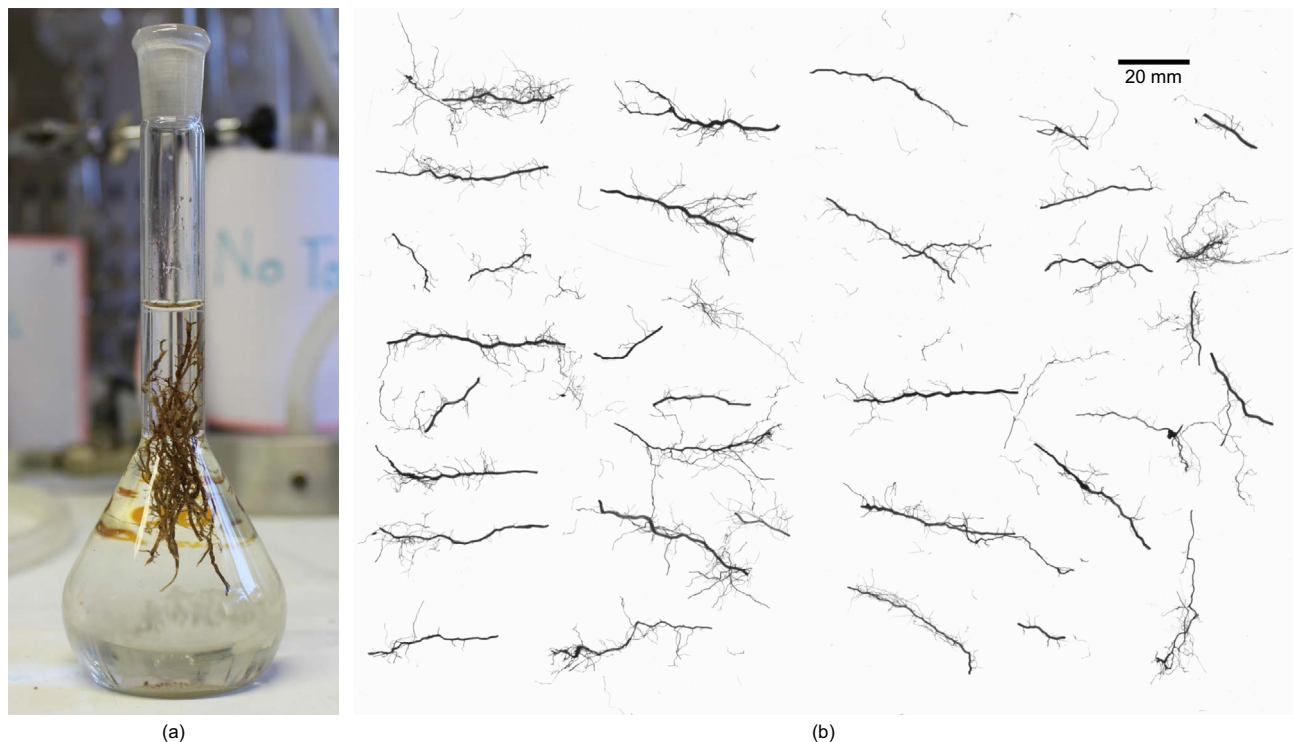


Fig. 4. Roots volume assessment by: (a) pycnometry; (b) WinRHIZO acquired image

Saturated water permeability

One bare and one vegetated soil samples were tested in large oedometer cells (150 mm dia., 70 mm high, Fig. 2(b)). The soil was compacted (initial state in Table 1), and plants were left to grow for 3 months in the equipment. Samples were kept saturated and under controlled hydraulic gradient conditions throughout all the loading/unloading steps. Saturation was assumed to be reached in oedometer samples when the pressure–volume controller observed steady-state outflow from the drainage line. The vertical stresses imposed were: 10/20/40/90/130 kPa on loading and 90/50/10 kPa on unloading. For each loading step, the permeability was evaluated using consolidation theory. Void ratios corresponding to the different loading steps were calculated, knowing the vertical strains that occurred.

A permeameter (initial state in Table 1, Fig. 2(c)) was used to perform a falling head permeability test (ASTM, 2016). Before testing, the specimen was fully saturated under a controlled hydraulic gradient. Saturation was checked by measuring the water pressure at three points along the height of the sample placed into the permeameter. Then, the void ratio was evaluated, after each test, with the paraffin technique already described above.

METHODS FOR OBSERVATIONS AT THE MICROSCOPIC SCALE

It is assumed that roots affect soil structure according to different processes (Carminati *et al.*, 2013; Ng *et al.*, 2014; Scholl *et al.*, 2014; Shao *et al.*, 2017; Lu *et al.*, 2020): the growth of roots induces soil fissuring, and those fissures can generate increasing gaps between roots and soil matrix, if a concurrent shrinkage of soil and roots occurs during the drying phase. Moreover, the decay of roots leaves fissures unclogged, with a consequent channel formation, which increases pores connectivity. At the root's micro-scale, root hairs (i.e. roots with sizes of approximately 10 μm) can clog micro-pores and release fluids (mucilage) that stick soil particles together and avoid clay aggregate swelling/shrinking.

The pore domains involved in these phenomena correspond well to the porosity ranges that can be investigated by the two techniques used: mercury intrusion porosimetry (MIP) and X-ray micro-tomography. As samples for MIP must be cubes of 10 mm side at maximum, they were systematically extracted near root tips and hairs, as shown in Fig. 5, and freeze-dried after X-ray micro-tomography to explore the effect of the plant-released mucilage and microscopical roots on soil fabric. No attempt was made to include thicker or massive systems of roots within MIP subsamples, as a representative volume of roots (i.e. having a root volume ratio close to that observed at larger scales, $R_v = 0.015$) should be 15 mm³, which is rather challenging to control. Root tips and hairs are responsible for producing mucilage around them, hence creating the so-called rhizosphere. The aim of MIP tests was to explore this indirect effect of roots (mucilage-induced void occlusion) on soil PSD. Moreover, as the smallest roots have diameters of about 10 μm , a relevant volume of pores explored by MIP (i.e. below 10 μm) could not be physically intruded by roots. The representativity of the MIP subsample's sizes will be discussed in the results, while the representativity of the X-ray subsamples was deemed to be satisfactory when reaching R_v values similar to those observed in the oedometer, retention and permeameter tests.

X-ray micro-tomography

X-ray micro-tomography scans were carried out on two bare and five vegetated soil samples, each containing an individual *Cynodon dactylon*. These samples were extracted from pots when the soil was at different hydraulic states (Table 2). A MicroCT scanner (V|Tome|X s 240, GE Sensing & Inspection Technologies GmbH, Germany) allowed a cone beam computed tomography (CT) at 170 kVp/250 mA to reconstruct image stacks with an isometric voxel size equal to 36 μm . Eight-bit image processing was done by *ImageJ* (Fiji) (Schindelin *et al.*, 2012). A region of interest (ROI) was defined, including 85% of the samples' total volume, to avoid their irregularities at the external boundaries. From the grey

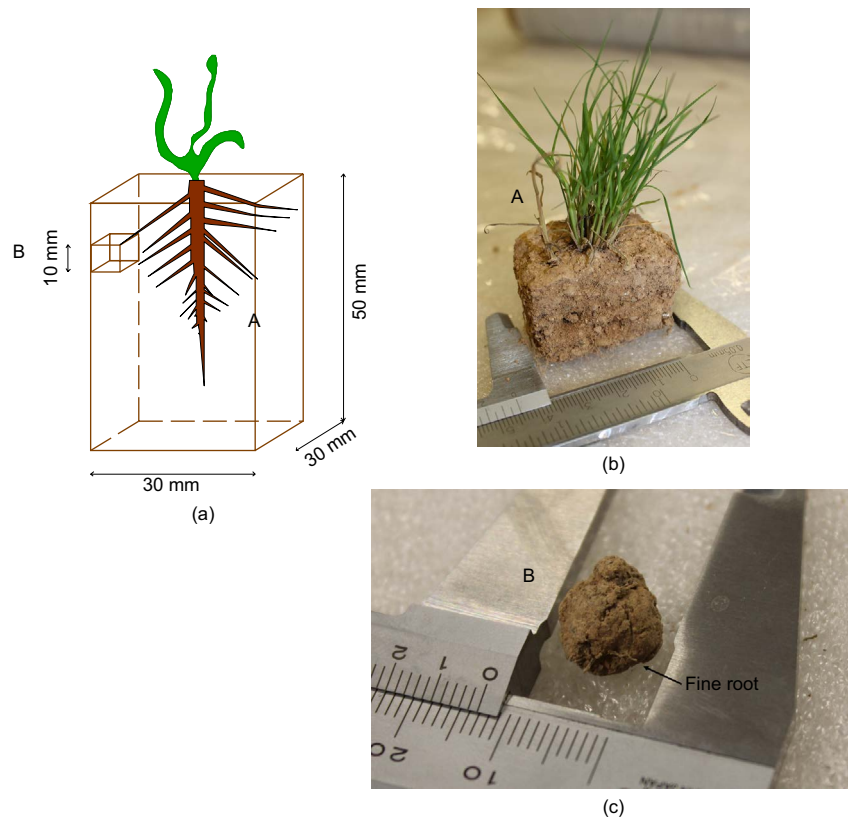


Fig. 5. (a) Schematic sample scanned by X-ray micro-tomography (A) and tested by MIP (B); (b) and (c) pictures of soil samples

Table 2. Testing state for X-ray micro-computed tomography and MIP samples

Soil initial state	Veg 1	Veg 2	Veg 3	Veg 4	Veg 5	Bare 1	Bare 2	Bare 3	Bare 4
Water ratio, $e_w = \rho_s w / \rho_w$: –	0.33	0.13	0.23	0.61	0.47	0.33	0.61	0.27	0.52
Matric suction, s : kPa	40	3050	420	0.5	3	110	0.7	310	4

value histogram (Fig. 6(a)) of each image stack (Fig. 6(b)), it was possible to identify the grey intensity peak corresponding to the roots and to isolate them by setting a threshold between 60 and 80 (see binarised figure in Fig. 7(a)). These values were confirmed by calculating the CT grey value of water, lignin and cellulose (main components of the roots) according to the equation (Hounsfield, 1972):

$$CT_{\text{grey val.}-12 \text{ bit}} = 1000 \left[1 + \left(\frac{\mu - \mu_w}{\mu_w} \right) \right] \quad (3)$$

Parameters μ_w and μ are, respectively, the linear attenuation coefficients of the water and a general material. These coefficients were found in the XCOM database (Berger *et al.*, 2010). More information on the calculations of the CT grey values is provided by Fraccica (2019).

The binarised stack generated still included roots and pores filled with water: to isolate roots only, it was treated, through a semi-automated procedure, with the Local Thickness plugin (Dougherty, 2007) to remove noise, pores filled by water and pores' borders with thicknesses below 4 pixels. The volume of roots was estimated based on the resulting image stacks (Fig. 7(b)) by the *ImageJ* plugin *MorphoLibJ* (Legland *et al.*, 2016), as Fraccica (2019) observed good agreement between X-ray, pycnometry and WinRhizo root volume measurements and relative errors in terms of porosity $(R_{V,X\text{-ray}} - R_{V,\text{pycnometry}}) / n_0$ (where n_0 is the initial soil porosity) and $R_{V,X\text{-ray}}$ and $R_{V,\text{pycnometry}}$ are the root volume ratios evaluated by X-ray and pycnometry) below 1%. Solid particles were isolated (black

pixels in Fig. 7(c)) from the other phases through an automatic threshold operation using *ImageJ*, based on the histogram of the entire image stack's grey values and the *Default* method. Void ratios calculated in bare soil X-ray images were comparable with those evaluated by paraffin tests. The threshold point was always close to 120 (Fig. 6(a)). The image stack containing the roots phase (previously shown in Fig. 7(b)) was then superimposed on Fig. 7(c) to include roots in the sample again and leave the voids in white pixels (Fig. 7(d)). A PSD was assessed for each scanned sample. The analysis was done on binarised images (see Fig. 7(d)), with the *ImageJ* plugin *Pore size distribution* (Münch & Lorenz, 2008). The tool can perform a three-dimensional (3D) simulation of mercury intrusion within the (white) voxels representing voids and evaluate the volume occupied for each class of pore diameter (Fig. 7(e)). The tool described allowed pores with a minimum diameter of 37 μm to be investigated, which is the resolution of the X-ray micro-tomography. Three-dimensional images of roots and voids (Fig. 7(f)) were also rendered to provide a first qualitative insight into the distribution of large pores and fissures around the plants.

Extension to the micro-pores range

Five vegetated and four bare soil cubes with a size smaller than 10 mm were trimmed from the samples previously scanned by X-ray micro-tomography (at hydraulic states expressed in Table 2) and used to perform MIP (Fig. 5).

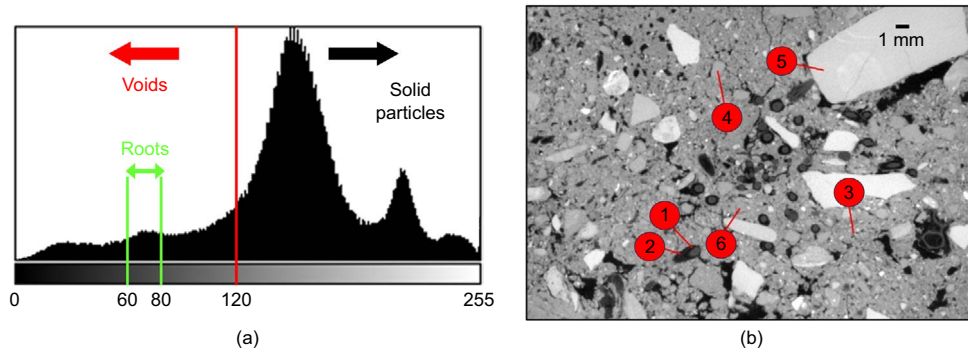


Fig. 6. (a) X-ray eight-bit stack histogram of the region of interest (ROI), with threshold values to distinguish the desired volumes. (b) Example of an X-ray section of the ROI. Different objects observed: 1, root cross-section; 2, fissure generated by root growth; 3, pre-existing pores; 4, sand grain; 5, gravel; 6, silty and clayey matrix

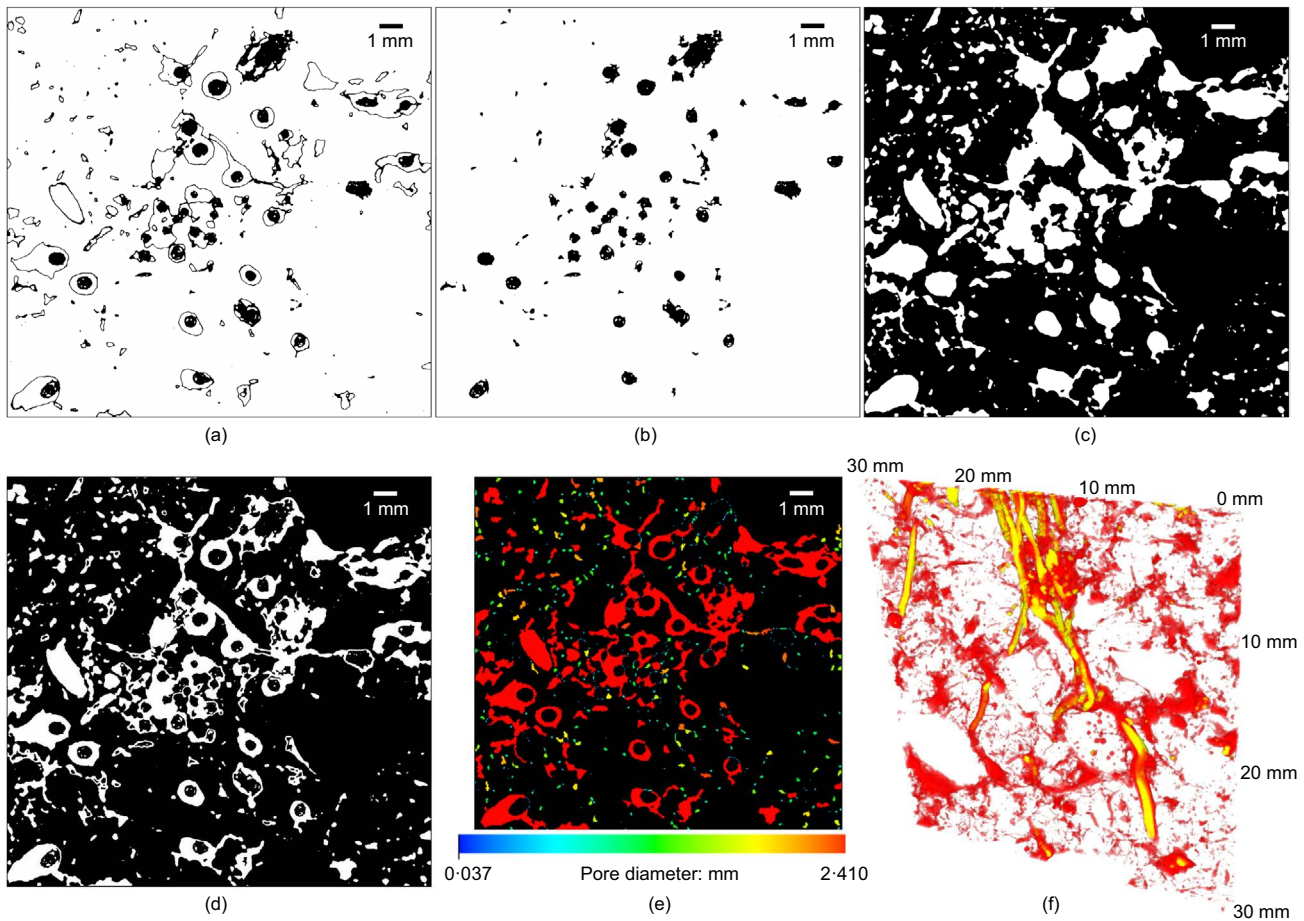


Fig. 7. (a) Binarised image with a threshold between 60 and 80: fissures and pores boundaries are still visible. (b) Roots phase wholly isolated from other soil phases. Different objects observed: 1, root cross-section; 2, fissure generated by root growth; 3, pre-existing pores. (c) Automatic threshold to separate solid particles (black pixels) from the remaining phases. (d) Superimposition of (b) on (c) to include the root phase within the image (only voids remained in white). (e) Individuation of the different pore diameter classes. (f) Three-dimensional rendering of roots (yellow) and pores (red) within a vertical section of an X-ray image stack

Before testing, all the cubes were freeze-dried to remove pore water, minimising soil shrinkage (Romero, 1999). Tests were carried out in a porosimeter (AutoPore IV 9500, Micromeritics Instrument Corporation, Norcross, GA, USA). This technique allowed investigating of pore sizes from 7 nm to 400 μm .

Evaluation of the pore size distribution

The PSD represents the cumulative void ratio occupied up to a given pore diameter, starting from the maximum diameter observed. For X-ray micro-tomography, the

cumulative void ratio for a given entrance pore size x^* was calculated by

$$e_{\text{cum}}^{(x^*)} = \frac{\sum_{x=x_{\text{max}}}^{x^*} V_{\text{void}}^{(x)}}{V_{\text{solid}}} \quad (4)$$

where $V_{\text{void}}^{(x)}$ is the volume of voids (white pixels in Fig. 7(d)) occupied by a generic entrance pore size x , and V_{solid} is the volume of solid particles, calculated as the total volume of the black pixels in Fig. 7(d). In the case of the results given by MIP, the cumulative void ratio is evaluated directly from

mercury intruded volume, as explained by Romero *et al.* (1999).

PHENOMENOLOGICAL SCALE RESULTS

Water retention properties

Results are presented in terms of water ratio ($e_w = \rho_s w / \rho_w = S_r e$) because they are easily comparable with void ratio variations since the two variables coincide at $S_r = 1$.

Data were fitted using the model of Romero & Vaunat (2000), based on a modified form of the van Genuchten (1980) expression:

$$S_r = C(s) \left[\frac{1}{1 + (\alpha s)^n} \right]^m; e_w = S_r e(s); C(s) = 1 - \frac{\ln[1 + (s/a)]}{\ln(2)} \quad (5)$$

where s is suction; S_r is the degree of saturation; $e(s)$ is the variation of void ratio as a function of suction; and $C(s)$ is a function used for adjusting the curve in the higher suction range. Parameter $1/\alpha$ is linked to the air-entry value, n to the slope of the curve, and m to the residual degree of saturation. Parameter a is considered to be the maximum suction for $S_r = 0$. Calibrated parameters are summarised in Table 3.

A comparison of the measurements obtained and the main drying curves fitted for bare and vegetated samples is presented in Fig. 8. The measurements carried out in the fallow soil were in good correlation with the respectively fitted curve, whereas correlation slightly decreased for the case of vegetated soil. Indeed, measurements belong to soil pots in which different values of root volume ratio R_v were observed ($R_v = V_{\text{roots}}/V_{\text{tot}}$, according to Fig. 3). The range of root volume ratios found in the pots is shown in the plot.

Table 3. Parameters calibrated for soil water retention curves

	α : MPa ⁻¹	n	m	a : MPa
Bare soil, $e = 0.67$	0.53	0.28	1.77	200
Root-permeated soil, $e = 0.67$	0.15	0.19	2.38	200

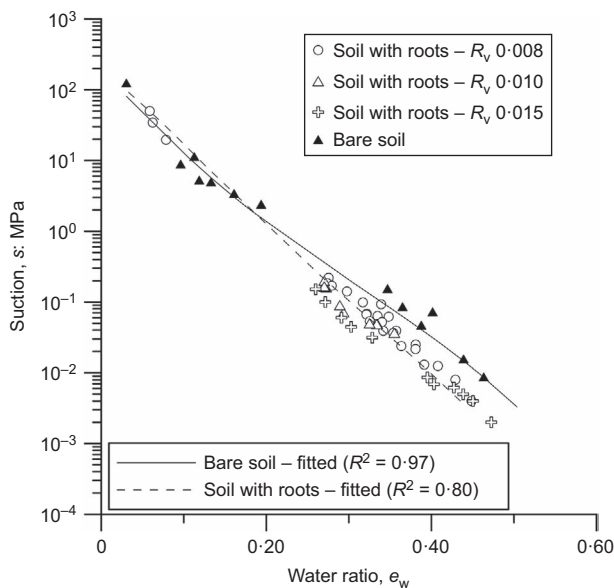


Fig. 8. Main drying curves for bare and root-permeated soil

Furthermore, it was observed that the air-entry value and the retention capacity of the vegetated soil have decreased. Moreover, this phenomenon is accentuated as R_v increases. Finally, all the retention measurements converge at a suction close to 3 MPa. A similar trend, in terms of degree of saturation, was observed by Fraccica *et al.* (2022) for laboratory and in situ measurements on the same soil as this study.

Volume evolution on drying

The results, in terms of void ratio, have been interpreted with the laws expressing volumetric deformations during suction changes proposed by Alonso *et al.* (1990):

$$de_{vs}^{el} = \frac{\kappa_s}{(1+e)} \frac{ds}{(s + p_{at})}, s < s_0 \quad (6)$$

$$de_{vs}^{pl} = \frac{\lambda_s}{(1+e)} \frac{ds_0}{(s_0 + p_{at})} \quad (7)$$

The parameters κ_s and λ_s are the stiffness values for suction changes along pre-yield and post-yield states; e is the void ratio before suction variation (ds); and p_{at} is the atmospheric pressure. The term s_0 indicates the maximum yield suction experienced by the material, above which irreversible strains are occurring. The initial maximum suction experienced by all the cubes was the as-compacted one, $s_0 = 40$ kPa (Table 1).

Void ratios measured by paraffin tests and shrinkage curves fitted by equations (6) and (7) are shown in Fig. 9. Vegetated soil samples remained at higher void ratios than the bare soil throughout the drying path. Similar results were observed by Milleret *et al.* (2009). Although a trend suggesting an increase in void ratio with increasing R_v was observed, it was decided not to implement this effect in equations (6) and (7) for several reasons: (a) the values of R_v observed in the void ratio measurements are not comparable with those observed in other tests; (b) the total void ratio couples different chemo-mechanical effects of vegetation at different pore size scales, as described above; and (c) given the small variations in total void ratios at a given suction, the effects of vegetation on water retention and soil permeability should be

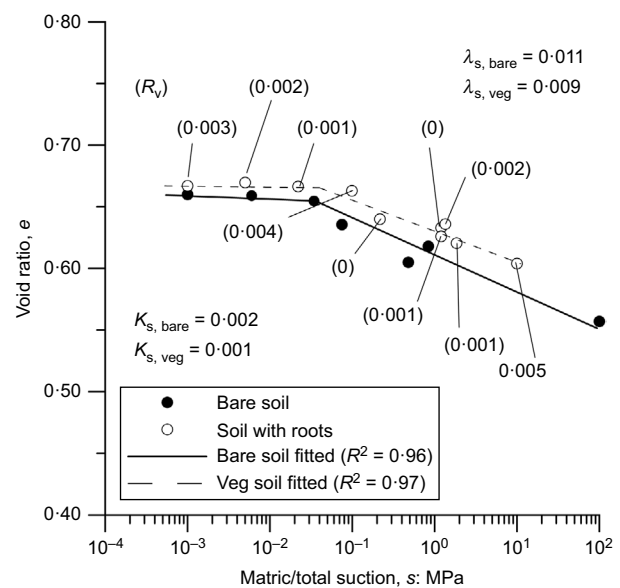


Fig. 9. Observed and predicted volumetric deformations, expressed in terms of void ratio along suction increase. Root volume ratio values are plotted together with the corresponding points

attributed primarily to variations in the size distribution of soil pores.

Water permeability

The values of water permeability were plotted as a function of the final void ratios (Fig. 10): results obtained by Oorthuis *et al.* (2018) for the same bare soil as this study are also included. The corresponding compaction vertical stresses ($\sigma_{v,0}$) and the root volume ratios (R_v) are also included.

The presence of vegetation made the water permeability increase, although differences between bare and vegetated soil became smaller at lower void ratios. These results agree

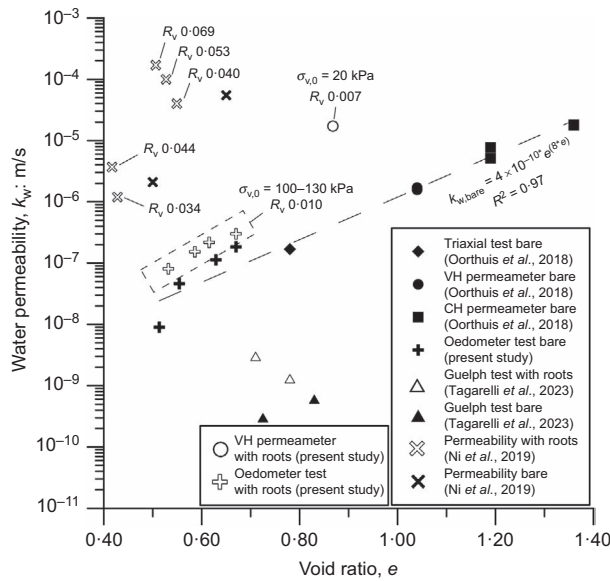


Fig. 10. Water permeability for bare and vegetated specimens evaluated for different void ratios and with different laboratory tests. Dashed line indicates the best-fit tendency. Void ratios from Ni *et al.* (2019) calculated according to the model proposed by the same paper

with other authors (Ng *et al.*, 2014; Vergani & Graf, 2016) who found that the hydraulic behaviour of a root-permeated soil depends on the initial compaction state, apart from roots' features and their evolution in time. This happens since the initial void ratio can limit or constrain the growth of roots in vertical and/or lateral directions (Daddow & Warrington, 1983; Correa *et al.*, 2019). Moreover, the stress state induced within soil upon compaction is not completely erased after unloading, posing an additional constraint to the energy that roots should exert to displace grains, during their development. These effects are behind the capacity of roots to create new macro-pores and penetrate the soil, with clear indirect consequences on soil hydraulic behaviour. Similar trends of increasing permeability, although in different soil types, have been observed in the literature (see Fig. 10): for instance, Ni *et al.* (2019) and Tagarelli *et al.* (2023) observed increases in permeability of more than one order of magnitude, at a given void ratio, when vegetation grew in a clayey sand (field and laboratory measurements) and in a clay (field measurements), respectively. In addition to the comparison with literature data, the results of this study in terms of permeability are consistent with the increase in porosity and decrease in water retention capacity. More measurements are, anyway, necessary to deepen the modelling capacity of permeability changes as a function of vegetation features.

RESULTS AT THE MICRO-SCALE

X-ray micro-tomography allowed the root architecture and volume within the soil samples to be evaluated. Fig. 11 represents the R_v distribution along X-ray slices (sample depth) of the five vegetated soil samples used for microstructural investigation in the study. An average R_v of the whole sample value was finally taken, for each sample, for further analyses and modelling. As it is possible to observe, the root distribution is quite homogeneous along the depth, except in the 'veg 5' sample (see Table 2). In this last case, a higher root content was observed at its contact with the plant's external part.

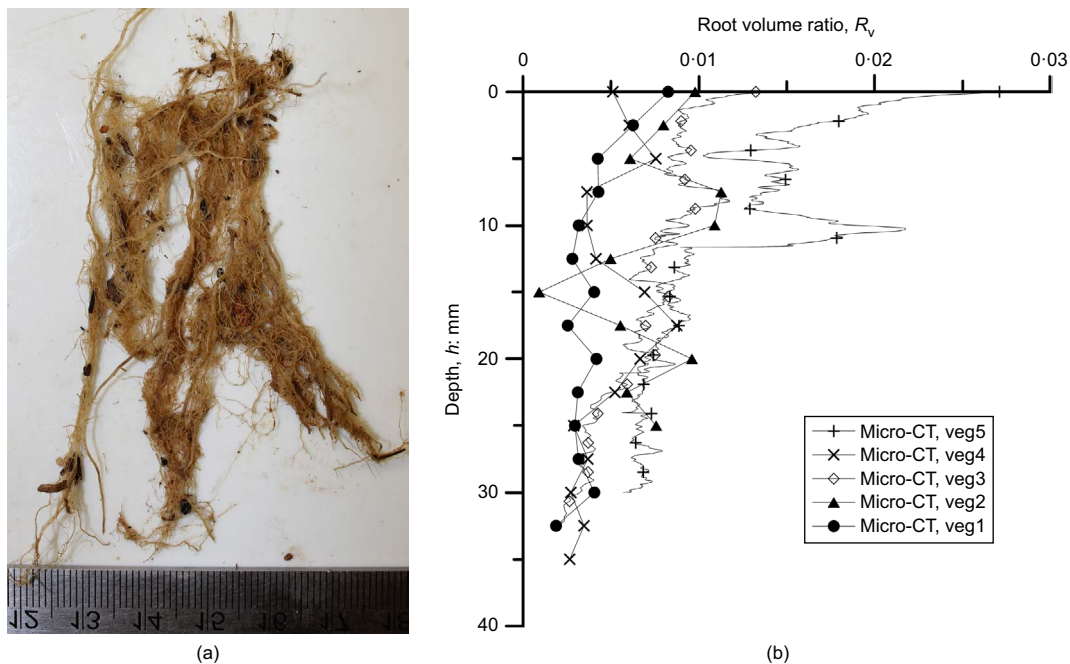


Fig. 11. (a) Example of root architecture after removal from the soil sample 'micro-CT, veg 5' and (b) root distribution along soil samples' depth obtained by X-ray micro-tomography

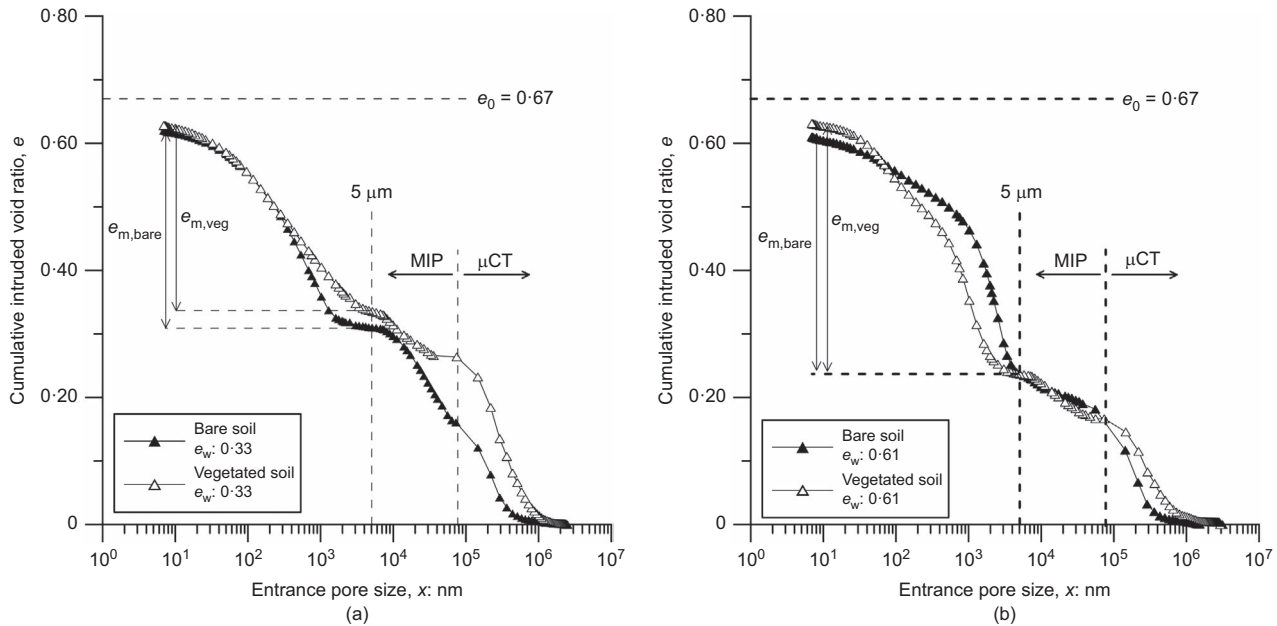


Fig. 12. Cumulative PSDs for bare and vegetated samples at two different water ratios: (a) $e_w = 0.33$; (b) $e_w = 0.61$ (note: μCT , micro-computed tomography)

The results obtained from the two microscopic techniques have been joined to make the upper end of the X-ray micro-tomography PSD curve coincide with the lower end of the MIP PSD curve. In Figs 12(a) and 12(b), examples of the combined PSD curves for bare and root-permeated soils at water ratios $e_w = 0.33$ and 0.61 are proposed. Even if a slight increase in the cumulated void ratio was observed in the vegetated soils, the most significant changes were observed in the vegetated PSD shape at $e_w = 0.33$, particularly in the higher range of pore sizes.

Results will be further presented as PSD derivatives for better individuating the soil's pore modes: in Figs 13(a) and 13(b), an overview of all the results is provided, where a three-modal porosity distribution can be distinguished. Pore sizes below $5 \mu\text{m}$ correspond to the clay aggregation range, which cannot be physically penetrated by roots or root hairs (Lamont, 1983; Wulfssohn & Nyengaard, 1999). Comparing Figs 13(a) and 13(b), one can note that, upon decrease of the water ratio e_w , the diameter at the peak (mode) of the clay aggregates in the bare soil undergoes a much larger shift towards smaller pore diameters (from 2.5 to $0.3 \mu\text{m}$) than the vegetated one (from 1 to $0.5 \mu\text{m}$); such dramatic changes cannot be ascribed to sample variability. As a result, clay aggregates experienced lower shrinkage in the vegetated samples, affecting the volume change behaviour observed at the phenomenological scale (Fig. 9). This phenomenon may be ascribed to the chemical clogging of pores, linked to root's mucilage production that increases upon drying, solidifying and bonding soil particles and root tips (Watt *et al.*, 1994).

In the pore size range from 5 to $100 \mu\text{m}$ (Figs 13(a) and 13(b)), pore volume density is lower in the vegetated samples; in this range of porosity, fine roots and root hairs can penetrate and clog voids (Koebernick *et al.*, 2017). Indeed, these root types (with diameters lower than $100 \mu\text{m}$) were detected by WinRhizo image analyses. However, the estimated volume of this root diameter class was only 0.7% of the total root volume, in average, thus explaining how larger root and pore sizes controlled the soil's hydraulic behaviour. Indeed, above $100 \mu\text{m}$ an opposite trend was observed: pore volume density increased in the rooted soil. In addition, given a fixed amount of roots, the volume of this range of

porosity is increasing as the soil dries out, being controlled by the concurrent shrinkage of soil and roots (Jones, 2007; Carminati *et al.*, 2013; Liu *et al.*, 2015). Image analyses detected roots of diameters between 100 and $1300 \mu\text{m}$ (root diameter peak at $500 \mu\text{m}$ (Fraccica, 2019)), the growth of which occurred in this last range of pores.

Although some general trends were traced, interesting phenomena were discovered by comparing curves at similar hydraulic states. For samples at a water ratio equal to $e_w = 0.61$ (Fig. 13(c)), it is possible to note that the volume of pores above $100 \mu\text{m}$ is almost equal between the two soils, even if the peak value corresponds to a larger pore size within the vegetated sample. This is in line with findings shown by Carminati *et al.* (2013), who showed a swelling capacity of roots to almost clog the fissures generated by their growth, when they absorb water or when soil is saturated. The slight increase of the macro-pore peak is consistent with the higher saturated permeability of the vegetated soil at the same void ratio (Fig. 10). In the diameter range below $5 \mu\text{m}$, some porosity is increasing in volume in the vegetated soil. Jointly to a partial pore clogging due to mucilage production, some pores in aggregates might have been generated due to other chemical activities in the soil. Indeed, high water contents increase soil nutrient mineralisation due to metabolic and enzymatic activities (Borken & Matzner, 2009; Manzoni *et al.*, 2012): this is strictly linked to plant nutrition (Singh & Schulze, 2015). Finally, after a drying period up to reaching a lower water ratio (Fig. 13(d), $e_w = 0.33$), most of the pores below $100 \mu\text{m}$ appear to be reduced in volume while, above $100 \mu\text{m}$, their volume increased.

Pore size distributions were used to evaluate soil water retention curves (SWRCs). Concerning the part of the PSD obtained by MIP, suction has been calculated from the mercury pressure values achieved for each intrusion step, according to Romero *et al.* (1999), assuming the mercury intrusion process (ruled by Washburn equation) to be similar to air intrusion (Young–Laplace's equation). Concerning the part of the PSD obtained by X-ray micro-tomography, suction was calculated by Young–Laplace's equation, considering that suction is generated only by capillary effects within this range of large porosities. The degree of saturation of the gas phase $S_{r,g}$ was calculated as the ratio between the

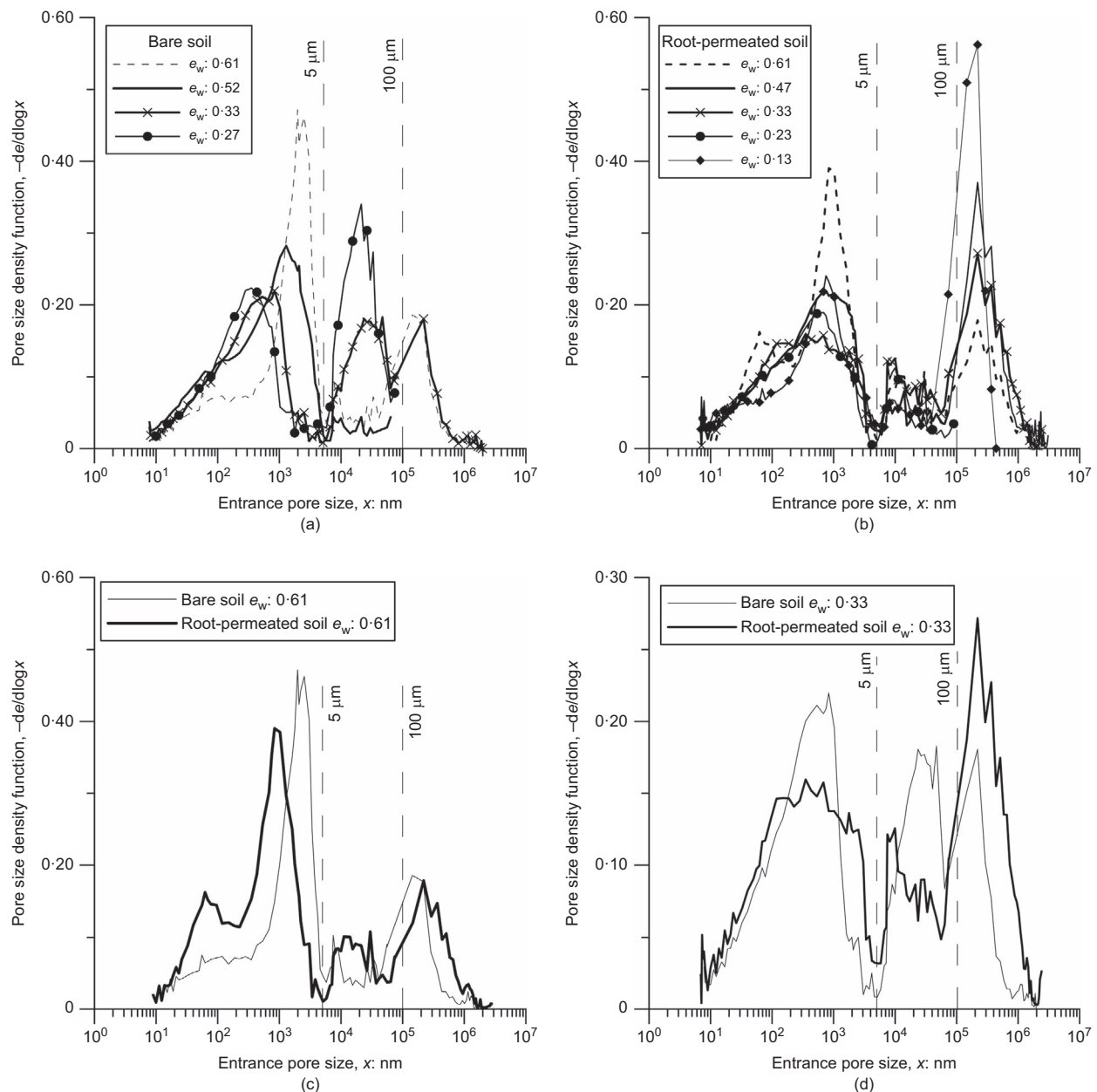


Fig. 13. PSD functions for: (a) all the bare samples; (b) all the root-permeated samples; (c) bare and root-permeated samples close to saturation ($e_w = 0.61$); and (d) after a drying phase ($e_w = 0.33$)

cumulative void ratio at a given pore diameter and the total cumulative void ratio (Romero *et al.*, 1999). The water phase degree of saturation $S_{r,w}$ was calculated from the gas phase degree of saturation ($S_{r,g} + S_{r,w} = 1$), and then the water ratio was calculated (Romero *et al.*, 1999). Two curves obtained from the bare and vegetated samples at a water ratio $e_w = 0.33$ (Fig. 13(d)) were plotted jointly with soil retention measurements (Fig. 14).

In both cases, water retention curves predicted from PSD agreed with the measured values and highlighted the effect of the fabric changes on the macroscopic behaviour. It is worth noting that the vegetated curve is predicting well the measurements with a similar root volume ratio ($R_v = 0.008$). Moreover, it is possible to note how fissures and macro-pores affect the air-entry value and the retention capacity in the low-suction range of the vegetated soil. Subsequently, the decrease in pore volume within the clay aggregates progressively controls the macroscopic behaviour and slightly deviates the water retention curve towards the bare one.

DISCUSSION

Effects of roots on soil microstructure

Soil microstructure evolution on drying is described by the intra-aggregate void ratio e_m ; this variable is defined on the PSD curves (Fig. 12) as the difference between the final cumulative intruded void ratio and that reached at the pore size of $5 \mu\text{m}$. The microstructural variable is linked to the total void ratio by

$$e = e_m + e_M \quad (8)$$

where e_M is the inter-aggregate void ratio, defined in this study as the cumulative void ratio of all the pores larger than $5 \mu\text{m}$.

Intra-aggregate void ratios from MIP and total void ratios from paraffin tests are plotted in Fig. 15, as a function of the water ratio at which samples were tested. For the bare soil, the swelling and shrinkage reversible behaviour of clay aggregates is indicated by the parameter β with the following equation in line with the one proposed by

Romero *et al.* (2011):

$$e_m = e_m^* + \beta e_w \quad (9)$$

where e_m^* is the intra-aggregate void ratio at a water ratio $e_w = 0$ and β is a microstructural parameter; both parameters depend on the specific surface of clay particles (Romero, 2013). Parameters for bare soil were calibrated minimising the root-mean-squared error and they are $e_m^* = 0.25$ and $\beta = 0.24$; they agree with the results shown by Romero (2013) for similar kinds of soils.

In Fig. 15, root volume ratios are labelled close to each X-ray + MIP measurement carried out for vegetated samples. Notably, values evaluated within microscopical investigations are comparable with those inferred within samples

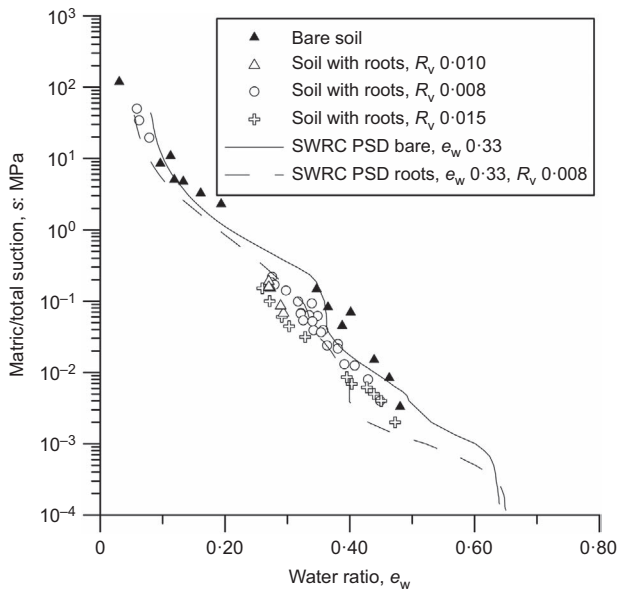


Fig. 14. Comparison of water retention measurements in bare and root-permeated soil and predicted curves from PSDs

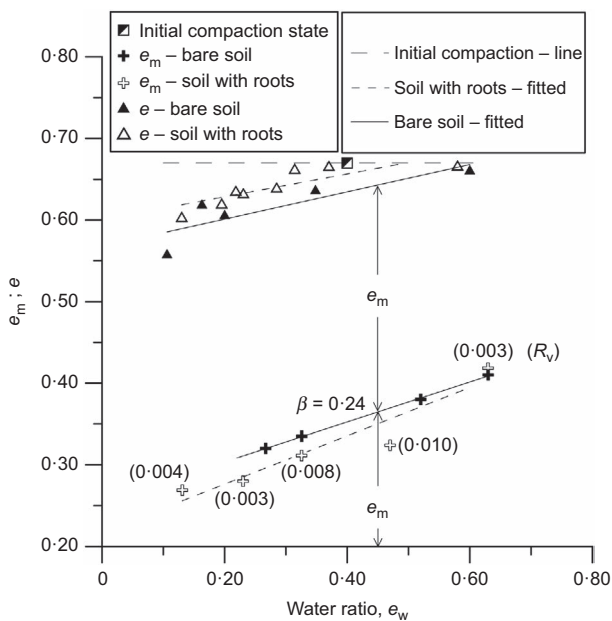


Fig. 15. Intra-aggregate and total void ratio evolutions with water ratio for bare and root-permeated specimens. Root volume ratios are labelled together with the corresponding points

for tests as larger scale (water retention and permeability), allowing for an easier up-scaling of the results from the micro- to the macro-scale. Despite the small presence of roots in the MIP subsamples, as previously mentioned, some changes in the corresponding PSD were observed: comparing Figs 13(a) and 13(b), a lower shrinkage evolution was observed in the vegetated soil by observing the diameter at the peak (mode) of the clay aggregates. Moreover, the different tendencies of the intruded micro-pore volume e_m in terms of the water ratio e_w (Fig. 15) suggest that a chemical effect of roots on soil occurred. However, the interpretation of the vegetated soil points in Fig. 15 is complex. Fissures and macro-pores volumes increase, while clay aggregates' pore volume reduces with drying, due to the chemical effects previously described. This is expressed by an increase in the macroscopic void ratio, e_m , and a consequent decrease of the intra-aggregate void ratio, e_m . To formulate this reduction, equation (9) was adapted as follows:

$$e_m = e_m^* + (\beta + \gamma^*)e_w - \alpha^* R_v(1 + e) ; v = 1 + e \quad (10)$$

in which γ^* and $\alpha^* R_v$ are the contributions added to the original model. R_v was multiplied to the specific volume v to normalise the root volume to the volume of solids, consistently with the formulation of the void ratio. The product $\alpha^* R_v$ formalises the amount of fissures and macro-pores generated by the normalised root volume R_v , at a given water ratio. The parameter γ^* controls the coupled effects of the presence of roots and water ratio variations as the micro-pores' volume changes with drying/wetting cycles and due to the plant's chemical activity. Higher values of the parameter α^* may either represent a soil that is more prone to be fissured or a plant with a root architecture that is generating a higher volume of fissures for a given R_v (i.e. plants with the same R_v can have many thin roots, as in this study, or few very thick roots, with potentially different effects on fissure generation). Higher values of γ^* may either represent a soil (or roots or both soil and roots) with a more pronounced shrinking/swelling behaviour upon water content changes or a root system that is producing higher quantities of mucilage. According to what was previously said, the values of α^* and γ^* depend on both soil and plant types. For the vegetated soil of this study, a calibration by minimising the mean squared error between measurements and predictions of e_m with equation (10) was carried out, while keeping e_m^* and β the same as for bare soil, to quantify parameters α^* and γ^* : their values were 1.35 and 0.028, respectively.

Prediction of the water permeability and the water retention behaviour

Equation (10) was used to predict measurements of permeability shown in Fig. 10.

The macropore volume e_m was evaluated for each measurement point, knowing the total void ratio e and the root volume ratio R_v observed in the samples. Soil water permeability was plotted as a function of the ratio between the macrostructural and the total void ratio. Although some further developments of the model can be envisaged, for example, to consider pore connectivity and size, as well as root diameter sizes and lengths, bare and vegetated soil samples were fairly represented by a unique correlation (Fig. 16).

The SWRCs were also predicted by adopting the framework developed by Dieudonné *et al.* (2013) for double-porosity soils given by equation (11), with the assumption that the water retention response of soil within the intra-aggregate's space is superimposed on the retention response

of the inter-aggregate's structural level. This assumption is formalised by the equation that links the water ratio values that fill micro- and macro-pores to suction:

$$e_w(s) = e_{wm} + e_{wM} = \frac{e_m}{e} \left[1 + \left(\frac{s}{s_0^{*(m)}} \right)^{n^{(m)}} \right]^{-m^{(m)}} + \frac{e_M}{e} \left[1 + \left(\frac{s}{s_0^{*(M)}} \right)^{n^{(M)}} \right]^{-m^{(M)}} \quad (11)$$

in which the two structural levels are expressed separately with a van Genuchten (1980) formulation. The quantities e_{wm} and e_{wM} are indicating the water ratio in the micro- and the macro-pores, respectively. The evolution of e_m as a function of the water ratio has been formalised in equations (9) and (10), while the evolution of e with drying is ruled by equations (6) and (7), considering the different values of κ_s and λ_s calibrated for bare and vegetated soils. n and m are model parameters, differentiated for the micro- and the macro-pores by the lower-case and upper-case exponent ((m) or (M), respectively). Finally, s_0^* is associated with the air-entry suction, and is expressed by the equations given by Dieudonné *et al.* (2013):

$$s_0^{*(m)} = \alpha_1^{(m)} \exp(-\alpha_2^{(m)} e_m) \quad (12)$$

$$s_0^{*(M)} = \alpha_1^{(M)} \exp(-\alpha_2^{(M)} \frac{e_M}{e}) \quad (13)$$

where α_1 and α_2 are model parameters.

The macrostructural and microstructural SWRC parameters of the retention curve were calibrated on the bare

soil water retention measurements. Their values are summarised in Table 4.

Once the retention curve parameters had been calibrated (bare SWRC in Fig. 17), they were used to validate the model and to predict the behaviour of vegetated soil samples presenting different root volume ratios. Equation (10) was used to predict the microstructural void ratio (e_m) evolution as a function of the R_v observed in the measurement pots ($R_v = 0.008$ and 0.015). The main drying curves calibrated on bare measurements and predicted by the model for the vegetated samples are shown in Fig. 17, jointly with the water retention measurements, observing a good agreement between them.

In particular, an increase in the root volume ratio and the different evolutions of the total and the microstructural void ratios observed in the vegetated samples allowed the prediction of an increase in macrostructural void ratio over total void ratio e_M/e . As expressed in equation (11), this increase induced a reduction of the air-entry value and of the retention capacity of the vegetated soil.

CONCLUDING REMARKS

The hydro-mechanical behaviour of root-permeated and bare compacted soils was investigated considering the phenomenological and microstructural scales. The soil was seeded by *Cynodon dactylon* and tested after 3 to 4 months of growth. Attention was paid to the initial soil state and stress and the hydro-mechanical paths followed during root growth and testing. The vegetated soil's water retention behaviour was evaluated by keeping plants in darkness and with null transpiration to observe the effects that the soil structure, modified by the roots, had on the macroscopic behaviour. Air-entry value, water retention capacity, saturated soil water permeability and shrinkage behaviour changed considerably due to the presence of vegetation. Although some phenomenological observations/trends proposed (SWRC and permeability) in this study are in line with other literature studies done on different types of soils and plants, this study focused on soil-roots interactions based on soil state (void ratio) and stress (suction) variables and the combination of MIP and X-ray micro-tomography, covering seven orders of magnitude of pore sizes. This made it possible to explain the changes observed at the phenomenological scale and predict the SWRC measurements using PSD curves obtained from experiments, which are totally independent and complementary to the former. The above-mentioned complementary techniques evidenced substantial changes at the macro-pores volume scale and physical-chemical effects at the micro-pores level of the soil-root system. The two phenomena were strongly conditioned by the soil hydraulic state and, in turn, affected the soil hydro-mechanical behaviour at the laboratory scale. The fissures reduced soil air-entry value and increased its saturated permeability, while the chemical changes affected the water retention curve at high suction ranges and contributed to reduced shrinkage upon drying. A state-variable-based model was also formulated to quantitatively correlate the normalised root volume to fabric changes in terms of cumulated volumes of different pore size ranges. This model was validated by predicting the soil water permeability and retention measurements. Although a

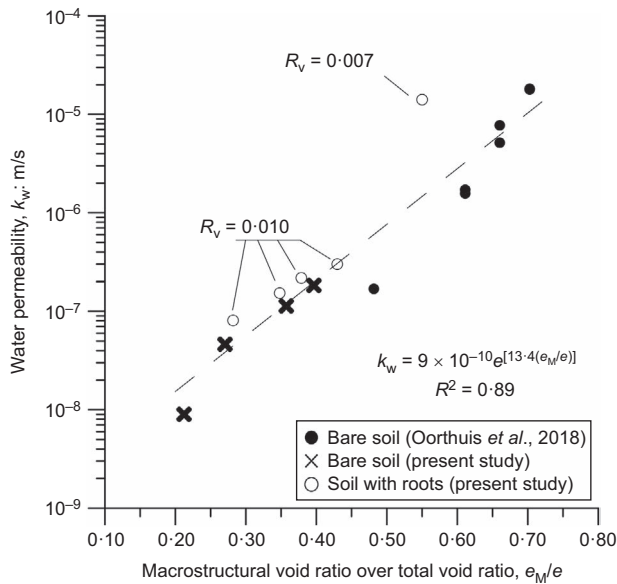


Fig. 16. Soil water permeability of bare and root-permeated soil as a function of the degree of macro-pores. Dashed line presents the best-fit tendency according to the equation shown in the plot

Table 4. Parameters calibrated for the double-porosity SWRC law formalised by Dieudonné *et al.* (2013)

$\alpha_1^{(m)}$: MPa	$\alpha_2^{(m)}$	$n^{(m)}$	$m^{(m)}$	$\alpha_1^{(M)}$: MPa	$\alpha_2^{(M)}$	$n^{(M)}$	$m^{(M)}$
1.0×10^{-1}	1.0×10^{-10}	59.0	4.2×10^{-3}	7.4×10^{-4}	7.9×10^{-3}	32.0	1.4×10^{-1}

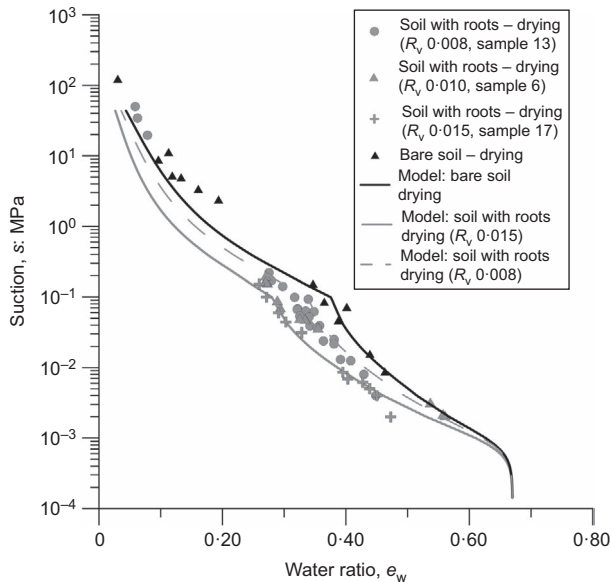


Fig. 17. Modelled soil water retention curves on main drying

direct upscaling to the site scale is not feasible, so far, the model could be helpful to evaluate how the SWRC and the permeability changes, if information on bare soil and vegetation root volume ratio only is available in a preliminary site study. Indeed, microstructural information on bare soil is largely available in the literature (e.g. Romero, 2013), for many types of soil. The model can then predict SWRC and permeability for a given range of expected R_v of the vegetation of the site, as done by Fraccica *et al.* (2023), for a real case of rainfall-induced landslide. Further improvements of the modelling framework will include pore connectivity, root diameter sizes, lengths, number and root decay indexes (for experiments on longer-living plants and submitted to drying/wetting cycles). Other limitations of the model are the lower efficiency in the prediction of permeability than in the prediction of retention capacity and the impossibility of using it if information on the bare soil microstructure is not available.

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NOTATION

e, e_m, e_M	total, micro-pores and macro-pores void ratios
e_w	water ratio defined as the volume of water to volume of solids
k_w	soil water permeability
L	root length
R_v	root volume ratio defined as the volume of roots to soil volume
S_r	degree of saturation
s	matric/total suction depending on measuring range/equipment (matric suction < 200 kPa)
$V_{tot}, V_{solid}, V_{roots}$	bulk, solid particles and roots' volumes
x	entrance pore size
a, n, m, a	soil water retention curve parameters
α^*, γ^*	parameters for micro-pores model
ϕ	root diameter

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